GOLDEN CRUST BAKERY

Inventory & Waste Intelligence Pipeline

A Plain-English Project Report

From Excel Spreadsheets to an Executive Power BI Dashboard: Building an End-to-End Data Pipeline to Find and Fix Bakery Waste

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September 2026

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1. Introduction — What This Project Is About

Golden Crust Bakery is a small, traditional bakery that sells fresh croissants, sourdough bread, chocolate pastries, baguettes, and cinnamon rolls every day. Like most small food businesses, the bakery has always tracked its work on paper and in simple spreadsheets: how much was baked each morning, how much was sold to customers, and how much was thrown away at closing time.

This project takes that everyday, familiar information and turns it into something the bakery's owners can actually use to make better decisions. In simple terms, it answers one core question: "Why are we throwing away so much bread, and what can we do about it?"

To answer that question properly, a small but complete data pipeline was built — a chain of tools that automatically collects the daily numbers, cleans them up, stores them safely, analyzes them, and finally displays the results as an easy-to-read dashboard. No part of this process requires the bakery staff to be data experts; once it is set up, it runs largely on its own.

2. The Business Problem

Every evening, Golden Crust Bakery's staff clear out the display shelves and throw away whatever bread and pastries did not sell. Over time, management noticed that a lot of product was ending up in the bin — but nobody could say for certain why.

Was too much being baked on certain days? Were customers simply not buying as much on some days of the week? Was the problem the same for every product, or did croissants behave differently from sourdough bread? Without clear numbers to look at, the bakery had no way of answering these questions with confidence, and every extra loaf thrown away was money lost.

This is a very common problem for small food businesses: the information needed to fix the issue already exists, scattered across production sheets, till receipts, and waste logs — but it is never brought together in one place where patterns can be seen.

3. The Goal of the Project

The goal of this project was to build an automated system that could:

- **Combine the data.** Bring together three separate records — what was baked, what was sold, and what was thrown away — into a single, organized source of truth.
- **Find the patterns.** Spot patterns in the numbers, such as certain days or certain products that waste more than others.
- **Give clear recommendations.** Turn those patterns into plain, practical advice that bakery managers can act on immediately, such as baking less on quiet days.

In short, the project turns raw daily records into a decision-making tool — the kind of system a much larger retail chain might use, built here at a small-bakery scale.

4. How the System Works — The Big Picture

Think of this project as a small factory line for information. Data goes in one end as simple spreadsheet rows, and a polished, colour-coded dashboard comes out the other end. Along the way, it passes through four stages:

Excel / Google Sheets ➔ Python (cleans & organizes data) ➔ SQL Database ➔ Power BI Dashboard

Here is what each stage means in everyday language:

Stage	Tool Used	What It Does, in Plain English
Data Entry	Microsoft Excel / Google Sheets	Where bakery staff write down daily numbers — easy for anyone to use, no special training needed.
Data Cleaning & Automation	Python (a programming language)	Automatically reads the spreadsheets, fixes formatting issues, and prepares the numbers for storage.
Data Storage	SQL Database (SQLite)	A well-organized digital filing cabinet that can hold months of records and answer questions about them instantly.
Reporting & Visuals	Power BI Desktop	Turns the stored numbers into charts, cards, and tables that management can read at a glance.

Each of these stages is explained step-by-step in the next section, along with real screenshots taken while the project was being built.

5. Step-by-Step: How the Pipeline Was Built

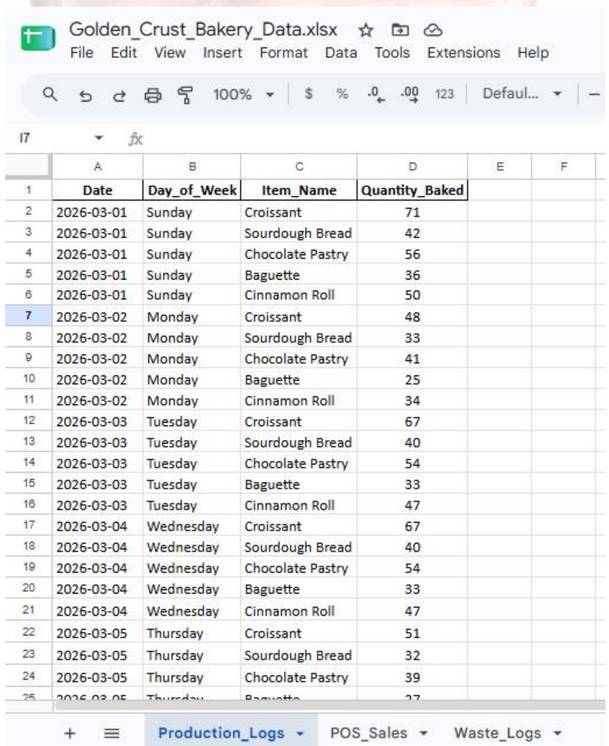
This section walks through exactly how the project came together, from the first blank spreadsheet to the finished dashboard. Each step includes a picture of the actual work.

5.1 Step 1 — Collecting the Raw Data

The project starts with a Google Sheets workbook named Golden_Crust_Bakery_Data.xlsx. This is the same kind of spreadsheet a bakery employee could fill in every day — nothing complicated. It has three tabs (like three pages in one notebook):

- **Production_Logs —** A daily record of how many of each item was baked.
- **POS_Sales —** A record of every individual customer sale, item by item.
- **Waste_Logs —** A record of how many of each item were left over and thrown away at closing.

Six months of records were used, covering March 2026 through August 2026, for five products: Croissant, Sourdough Bread, Chocolate Pastry, Baguette, and Cinnamon Roll.



	Date	Day_of_Week	Item_Name	Quantity_Baked
2	2026-03-01	Sunday	Croissant	71
3	2026-03-01	Sunday	Sourdough Bread	42
4	2026-03-01	Sunday	Chocolate Pastry	56
5	2026-03-01	Sunday	Baguette	36
6	2026-03-01	Sunday	Cinnamon Roll	50
7	2026-03-02	Monday	Croissant	48
8	2026-03-02	Monday	Sourdough Bread	33
9	2026-03-02	Monday	Chocolate Pastry	41
10	2026-03-02	Monday	Baguette	25
11	2026-03-02	Monday	Cinnamon Roll	34
12	2026-03-03	Tuesday	Croissant	67
13	2026-03-03	Tuesday	Sourdough Bread	40
14	2026-03-03	Tuesday	Chocolate Pastry	54
15	2026-03-03	Tuesday	Baguette	33
16	2026-03-03	Tuesday	Cinnamon Roll	47
17	2026-03-04	Wednesday	Croissant	67
18	2026-03-04	Wednesday	Sourdough Bread	40
19	2026-03-04	Wednesday	Chocolate Pastry	54
20	2026-03-04	Wednesday	Baguette	33
21	2026-03-04	Wednesday	Cinnamon Roll	47
22	2026-03-05	Thursday	Croissant	51
23	2026-03-05	Thursday	Sourdough Bread	32
24	2026-03-05	Thursday	Chocolate Pastry	39

The Production_Logs tab, showing daily baked quantities per item and day of the week.

5.2 Step 2 — Generating Realistic Sample Data with Python

To make sure the system would work with real-world patterns — not just random numbers — a Python script was written to simulate six months of genuine bakery activity. Python is a programming language that can follow instructions automatically, much faster and more reliably than doing the same work by hand.

The script was told to build in two realistic bakery behaviours on purpose, so the final analysis would have something meaningful to uncover:

- **A weekend rush —** Weekends (Friday to Sunday) see roughly 40% more customers than an average day, which is common for bakeries.
- **A midweek over-baking habit —** On Tuesdays and Wednesdays, staff tend to bake more than usual out of habit, even though fewer customers show up early in the week.

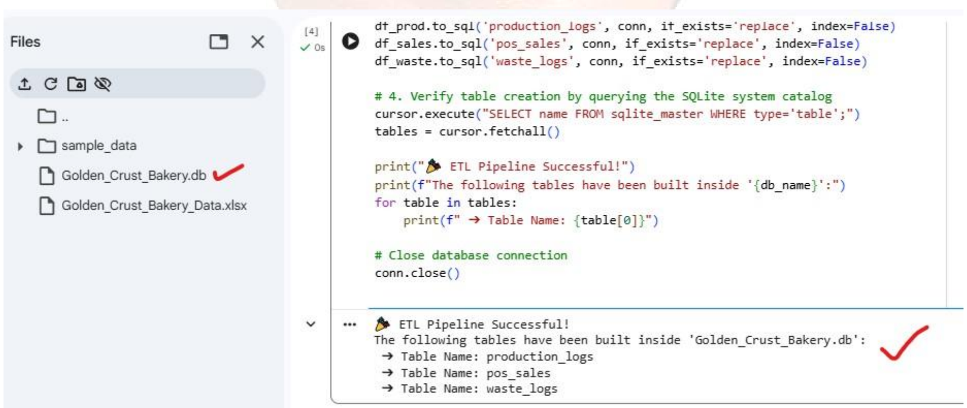
The script then calculated, for every single day and every product, how much was baked, how much sold, and how much was wasted (simply: baked minus sold), and saved everything neatly back into the spreadsheet.

The Python script running successfully, confirming the spreadsheet was generated with 920 production records, 2,760 sales transactions, and 920 waste records.

5.3 Step 3 — Moving Data into a Proper Database

A spreadsheet is fine for small amounts of information, but it becomes slow and unreliable once there are thousands of rows to search through. To solve this, a second Python script was used to move all the data out of the spreadsheet and into a proper database — a system built specifically for storing and quickly searching large amounts of information.

The database used here is called SQLite. It works like a set of neatly labelled, searchable tables rather than a single flat spreadsheet page, which makes it much faster and safer to work with as the data grows.

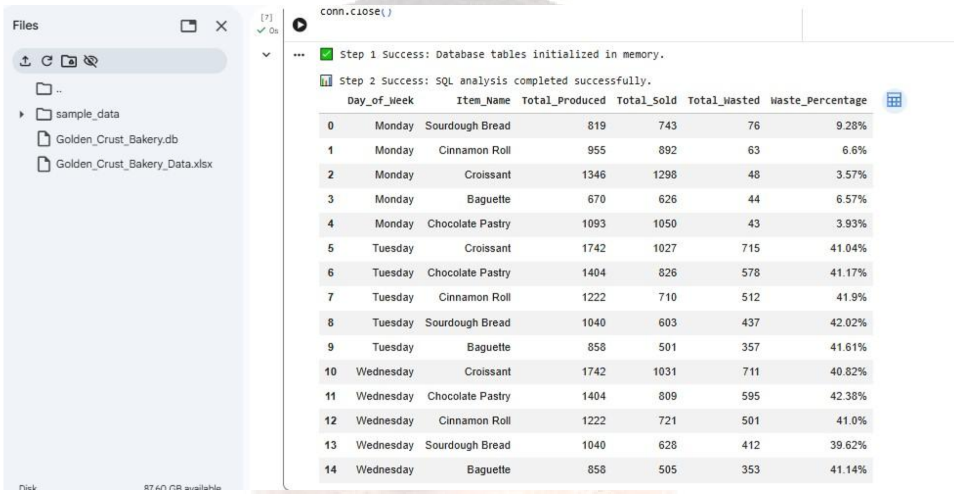


Confirmation that the three tables — production_logs, pos_sales, and waste_logs — were successfully created inside the Golden_Crust_Bakery.db database.

5.4 Step 4 — Asking the Database Smart Questions

With the data safely stored, the next step was to ask it useful questions using SQL — a language designed specifically for querying (asking questions of) databases. In plain terms, a query is simply a very precise question, such as: "For every day of the week and every product, how much was baked, how much was sold, and how much was thrown away — and what percentage of it went to waste?"

One technical challenge came up here: the sales records are very detailed (one row per customer purchase), while the baking and waste records are one row per day. Comparing them directly would have counted some numbers multiple times and produced incorrect totals. This was solved by first summarising all of a day's individual sales into a single daily total, and only then comparing it against the baking and waste figures — avoiding any inflated results.



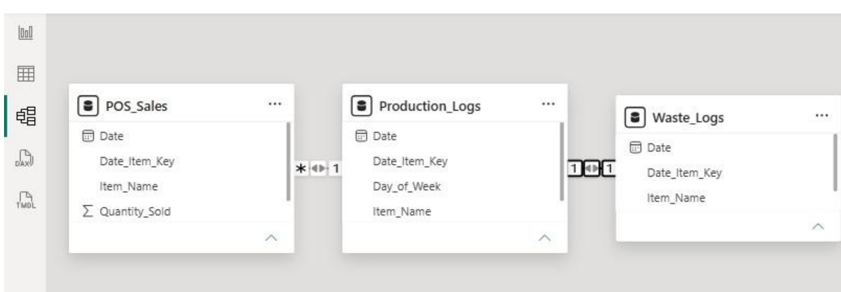
	Day_of_Week	Item_Name	Total_Produced	Total_Sold	Total_Wasted	Waste_Percentage
0	Monday	Sourdough Bread	819	743	76	9.28%
1	Monday	Cinnamon Roll	955	892	63	6.6%
2	Monday	Croissant	1346	1298	48	3.57%
3	Monday	Baguette	670	626	44	6.57%
4	Monday	Chocolate Pastry	1093	1050	43	3.93%
5	Tuesday	Croissant	1742	1027	715	41.04%
6	Tuesday	Chocolate Pastry	1404	826	578	41.17%
7	Tuesday	Cinnamon Roll	1222	710	512	41.9%
8	Tuesday	Sourdough Bread	1040	603	437	42.02%
9	Tuesday	Baguette	858	501	357	41.61%
10	Wednesday	Croissant	1742	1031	711	40.82%
11	Wednesday	Chocolate Pastry	1404	809	595	42.38%
12	Wednesday	Cinnamon Roll	1222	721	501	41.0%
13	Wednesday	Sourdough Bread	1040	628	412	39.62%
14	Wednesday	Baguette	858	505	353	41.14%

Query results showing, for each day and product, the total produced, total sold, total wasted, and the resulting waste percentage. Tuesday and Wednesday clearly stand out with waste percentages above 40%.

5.5 Step 5 — Building the Power BI Data Model

Power BI is the tool used to turn the database numbers into visual charts and dashboards. Before the charts could be built, the different tables (production, sales, and waste) had to be properly linked together, the same way you might link related pages in a binder so information can be cross-referenced.

The first attempt at linking the tables caused a common problem: the connections were ambiguous, meaning Power BI could not always tell which rows belonged together, which made some charts show the same repeated totals no matter what was selected. This was fixed by combining the Date and Item_Name fields into a single combined label (called a composite key) in every table, and rebuilding clean, one-directional and two-directional links between them.



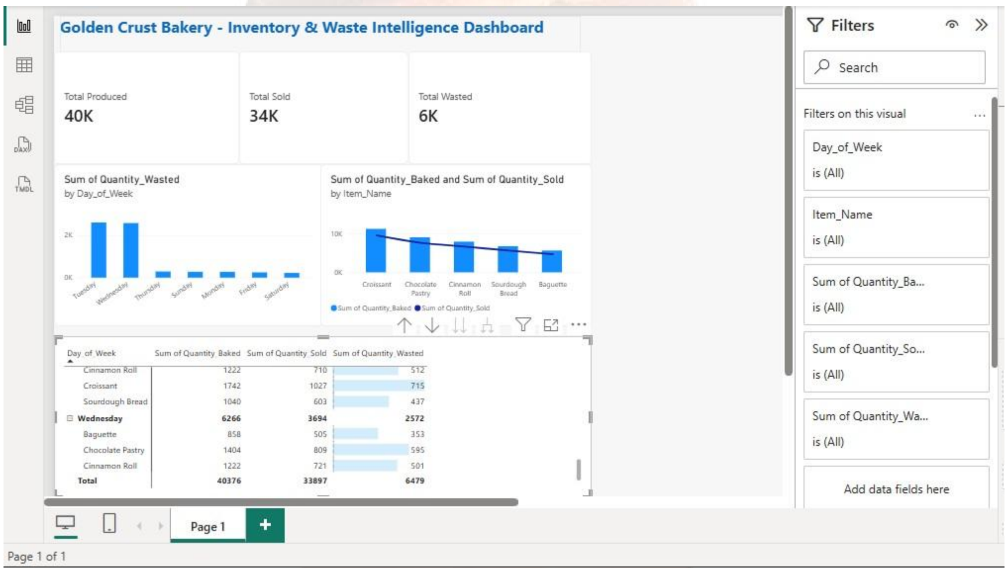
The corrected Power BI data model, showing clean, properly linked relationships between the Production_Logs, POS_Sales, and Waste_Logs tables via a shared Date_Item_Key.

5.6 Step 6 — Designing the Executive Dashboard

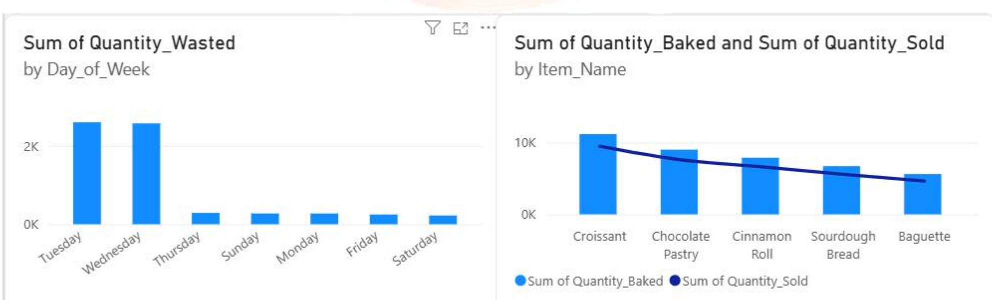
The final step was to design a dashboard — a single-screen summary that a busy bakery manager could glance at and immediately understand, without needing to read raw numbers or spreadsheets.

The finished dashboard includes:

- **Summary cards —** Three simple headline numbers at the top: total items produced, total items sold, and total items wasted.
- **Waste by day chart —** A bar chart showing exactly which days of the week waste the most product.
- **Production vs. sales chart —** A chart comparing how much of each product was baked versus how much actually sold, making mismatches easy to spot.
- **Detail table —** A detailed table with colour-coded bars so the worst-performing day-and-product combinations stand out visually.



The final executive dashboard: 40,000 items produced, 34,000 sold, and 6,000 wasted overall — with Tuesday and Wednesday standing out as the biggest problem days.



A closer look at two dashboard charts: waste is heavily concentrated on Tuesday and Wednesday (left), while Croissant is both the most-baked and most-wasted product overall (right).

Day_of_Week	Sum of Quantity_Baked	Sum of Quantity_Sold	Sum of Quantity_Wasted
Chocolate Pastry	1404	826	578
Cinnamon Roll	1222	710	512
Croissant	1742	1027	715
Sourdough Bread	1040	603	437
Wednesday	**6266**	**3694**	**2572**
Baguette	858	505	353
Total	**40376**	**33897**	**6479**

The detailed breakdown table for Wednesday, showing every product's baked, sold, and wasted quantities, with data bars flagging the worst offenders (Croissant and Chocolate Pastry).

6. What We Discovered — Key Insights

Once the dashboard was complete, three clear patterns emerged from six months of data:

Insight 1: Tuesdays and Wednesdays Are the Problem Days

Waste percentages jump to over 40% on Tuesdays and Wednesdays — meaning nearly half of everything baked those days goes unsold. This lines up exactly with the over-baking habit built into the data: staff keep baking at their usual pace even though far fewer customers come in early in the week.

Insight 2: Weekends Are Actually Under-Supplied

On the opposite end, Friday through Sunday show waste levels under 8%, with popular items like Sourdough Bread often selling out before the day is over. This means the bakery is likely losing potential sales on its busiest days simply because it is not baking enough.

Insight 3: Croissants Drive the Biggest Losses

Across the whole six-month period, Croissant is both the single most-produced and the single most-wasted item. Because it is baked in the highest volume every day, even a small percentage of waste on croissants adds up to a large number of wasted units overall.

Day of Week	Typical Waste Level	What It Means
Tuesday	~41%	Nearly half of what's baked is thrown away.
Wednesday	~40%	Same over-baking pattern as Tuesday.
Monday / Thursday / Friday	Low–Moderate	Baking roughly matches customer demand.
Saturday / Sunday	Under 8%	Popular items often sell out — demand may be under-met.

7. Recommendations for the Bakery

Based on these insights, three practical, low-cost changes are recommended:

- **1. Lower midweek baking by about 30%.** Since Tuesday and Wednesday demand is consistently lower, reducing production on those two days by around 30% would cut waste sharply while still leaving enough stock for customers.
- **2. Raise weekend baking by about 15%.** Because popular items sell out early on weekends, increasing production by around 15% on Fridays, Saturdays, and Sundays would let the bakery capture sales it is currently missing out on.
- **3. Repeat this process automatically, every week.** The same Python and SQL tools built for this project can be reused every week: each Sunday night, the previous week's sales and waste figures can be automatically processed to suggest baking quantities for the week ahead — turning this from a one-time report into an ongoing habit.

None of these changes require new equipment or additional staff — they are adjustments to existing daily baking quantities, guided by evidence instead of guesswork.

8. Conclusion

This project shows how a small, everyday business problem — too much bread going to waste — can be solved using the same kind of data pipeline that large companies rely on, scaled down to fit a single bakery.

By connecting simple spreadsheets to an automated Python process, a proper database, and a clear Power BI dashboard, scattered daily notes were turned into a single source of truth that anyone at the bakery can understand at a glance. The result is not just a set of charts, but a set of specific, actionable changes — baking less on Tuesdays and Wednesdays, baking more on weekends — that can directly reduce waste and protect the bakery's profit margin.

More broadly, this project demonstrates a complete, practical data workflow: collecting data, cleaning and automating it, storing it properly, analyzing it correctly, and presenting it clearly — the same core skills used in data analytics and business intelligence roles across many industries.

9. Glossary of Terms

For readers who are not familiar with technical terms, here is a plain-English explanation of the key words used in this report:

Term	Plain-English Meaning
Python	A programming language used to automatically read, clean, and organize data instead of doing it by hand.
ETL (Extract, Transform, Load)	The process of pulling data out of one place (a spreadsheet), cleaning it up, and loading it into another place (a database).
SQL	A language used to ask questions of a database, such as "how much was wasted on Tuesdays?"
SQLite	A lightweight, self-contained database used to store the bakery's records in an organized, searchable way.
CTE (Common Table Expression)	A way of asking a database to first summarise some data before comparing it to other data, to avoid incorrect totals.
Composite Key	Combining two pieces of information (like a date and a product name) into one label so records can be matched up correctly.
Power BI	A tool made by Microsoft that turns raw numbers into charts, graphs, and dashboards.
Dashboard	A single screen that summarises the most important numbers and charts so they can be understood at a glance.
Waste Percentage	The share of everything baked that ended up thrown away, calculated as wasted items divided by baked items.